ModelIt!

Teacher's Guide

What Happens When Wolves Disappear?

Modeling Trophic Cascades, Population Dynamics, and Ecosystem Stability

Quick Reference

Grade Level	Grade 10L1
NGSS Standard	HS-LS2-1, HS-LS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-1, HS-LS2-2

Use mathematical and/or computational representations to support explanations of factors that affect carrying capacity of ecosystems at different scales. Use mathematical representations to support and revise explanations based on evidence about factors affecting biodiversity and populations in ecosystems.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational models to represent factors affecting carrying capacity and predict population dynamics across multiple trophic levels.
Disciplinary Core Idea	**LS2.A Interdependent Relationships in Ecosystems / LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Ecosystems have carrying capacities that depend on biotic and abiotic factors. Trophic cascades demonstrate how interdependent relationships maintain or disrupt ecosystem functioning.
Crosscutting Concept	**Stability and Change** — Students analyze how the removal or addition of a keystone species pushes ecosystems between stable states and investigate the conditions required for ecosystem recovery.

CAST Testing Alignment

MULTIPLE CHOICE:

After wolves were removed from Yellowstone, elk populations increased by 300%. According to the trophic cascade model, which of the following would be the most likely SECOND-ORDER effect?

CONSTRUCTED RESPONSE:

Using your model, explain the complete trophic cascade that occurred when wolves were removed from Yellowstone. Trace the effects from predator removal through prey population increase, vegetation change, erosion, and river course alteration. Include at least three trophic levels in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

In 1995, one of the most famous experiments in ecological history began: wolves were reintroduced to Yellowstone National Park after a 70-year absence. What happened next stunned scientists — the wolves didn't just control elk populations, they transformed the entire landscape. Rivers moved. Forests regrew. Species that had vanished returned. This is the power of a trophic cascade: a change at the top of the food web that ripples through every level, reshaping the physical geography of the land itself.

The Yellowstone Experiment

Wolves were eliminated from Yellowstone by 1926 as part of federal predator control programs. Without wolves, elk populations grew unchecked from about 5,000 to over 20,000. The elk overgrazed willows, aspens, and cottonwoods — especially along rivers where they felt safe without predators. By the 1990s, riparian vegetation was devastated, riverbanks were eroding, and biodiversity had plummeted. In 1995, 31 wolves were reintroduced from Canada.

The Trophic Cascade

Within years of wolf reintroduction, elk began avoiding areas where they were vulnerable to wolf predation — especially river valleys and narrow gorges. This 'ecology of fear' allowed willows and aspens to regrow in these areas for the first time in decades. Vegetation recovery stabilized riverbanks, reducing erosion. Beavers returned (willows are their primary food), building dams that created ponds for fish, amphibians, and waterfowl. Songbird populations increased with the return of nesting habitat.

Carrying Capacity and Population Dynamics

Carrying capacity (K) is not fixed — it changes as the ecosystem changes. When wolves were absent, elk exceeded the vegetation carrying capacity, degraded the habitat, and lowered the carrying capacity further — a destructive feedback loop. Wolf reintroduction broke this loop by reducing elk numbers below the vegetation recovery threshold, which restored habitat, which increased carrying capacity for dozens of other species.

Keystone Species and Biodiversity

A keystone species has an effect on its ecosystem disproportionate to its population size. Wolves comprise a tiny fraction of Yellowstone's biomass but their presence or absence determines the structure of the entire ecosystem. Similarly, sea otters control sea urchin populations that would otherwise destroy kelp forests; starfish control mussel populations on rocky shores. Removing a keystone species doesn't just reduce biodiversity — it fundamentally reorganizes the ecosystem into a different, often degraded state.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify population size, food availability, predator pressure, habitat area, and extinction risk as the key components of the ecosystem dynamics model.

E **Establish Relationships**
Students determine that predator pressure controls population size, which determines food availability, which affects habitat area, and that the cascade from top predator to physical landscape demonstrates interconnected trophic levels.

V **Visualize & Model**
Students build a computational model showing how changes in predator pressure cascade through prey populations, vegetation, habitat quality, and biodiversity.

E **Evaluate Outcomes**
Students run wolves-present, wolves-removed, and wolf-reintroduction scenarios to observe trophic cascades and test whether degraded ecosystems can recover.

R **Revise & Extend**
Students add behavioral response, riparian vegetation, or species diversity count to model more complete ecosystem dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"When wolves disappeared from Yellowstone, the rivers literally moved. When wolves came back, the rivers returned. How is that possible?"*

Watch For

Show the famous 'How Wolves Changed Rivers' comparison photos. Let students try to explain the connection.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling something that might be the most surprising discovery in ecology — how one species can reshape an entire landscape."*

Watch For

Have students read objectives. Pre-teach 'trophic cascade' and 'keystone species' with visual food web.

Slide 3: Big Question

3 min

Say: *"Wolves eat elk. But somehow, removing wolves changed rivers, forests, songbirds, fish, and even soil erosion. What's the connection?"*

Watch For

Students attempt to draw the connection from wolves to rivers. Most will struggle — that's the point.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of an entire ecosystem to see how changes at the top cascade all the way to the bottom."*

Watch For

Preview LEVER steps. Emphasize this is about SYSTEMS thinking — everything is connected.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key players in this ecosystem? Which ones can wildlife managers actually control?"*

Watch For

Guide sorting. Predator pressure is the primary external lever. Discuss how everything else responds to it.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the cascade: No wolves means more elk. More elk means less vegetation. Less vegetation means... what?"*

Watch For

Build the full cascade chain step by step. Students add each link and predict the next.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's remove the wolves and watch what happens over 70 years. Then bring them back and see if we can undo the damage."*

Watch For

Run wolves-removed scenario first. Then run reintroduction. Compare timescales of degradation vs. recovery.

Slide 8: Discoveries

5 min

Say: *"It took 70 years without wolves to degrade Yellowstone. It took only 20 years with wolves to start recovery. What does that tell us?"*

Watch For

Discuss asymmetry: degradation is faster than recovery. Some damage may be irreversible. Connect to conservation urgency.

Slide 9: STEM Challenge

12 min

Say: *"A state agency wants to bring wolves back to a region where they've been gone for 80 years. Ranchers are furious. Ecologists say it's essential. Design the plan."*

Watch For

Groups create reintroduction plans balancing ecology, human concerns, corridors, and monitoring. Present and debate.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Predator Pressure
- **INTERNAL:** Population Size, Food Availability, Habitat Area, Extinction Risk

Relationships

- **Predator Pressure to Population Size:** NEGATIVE (-) — Higher predator pressure reduces prey population through direct predation and the ecology of fear (behavioral changes that reduce reproduction and survival). Wolf presence keeps elk numbers well below the vegetation carrying capacity.
- **Population Size to Food Availability:** NEGATIVE (-) — Larger prey populations consume more vegetation, reducing food availability for all herbivores. When elk populations exceed carrying capacity, they overgraze to the point where vegetation cannot recover.
- **Food Availability to Habitat Area:** POSITIVE (+) — Greater food availability means healthier, more abundant vegetation, which provides habitat for many species. Riparian vegetation stabilizes riverbanks, creates nesting sites, and supports the entire food web. When vegetation is destroyed by overgrazing, habitat area shrinks for all dependent species.

Key Discoveries

- Removing a top predator doesn't just increase prey numbers — it triggers a cascade that transforms the entire ecosystem, including vegetation, other species, and even physical geography
- The 'ecology of fear' may be as important as direct predation — wolves change WHERE elk graze, not just HOW MANY elk there are, allowing vegetation recovery in areas elk now avoid
- Ecosystem recovery after predator reintroduction is possible but slow — some changes (soil erosion, species loss) may take decades to reverse and some are irreversible
- Trophic cascades demonstrate that ecosystems are not simple food chains but complex interconnected webs where changes at any level produce unpredictable effects throughout

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Removing one species can't affect the whole ecosystem"	Trophic cascades demonstrate that a single keystone species can determine the structure of an entire ecosystem. Removing wolves from Yellowstone changed elk populations, vegetation, river courses, beaver populations, songbird habitat, and soil stability. The cascade model shows that ecosystems are tightly interconnected — a change at any level propagates throughout the system.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Nature will balance itself without predators"	Without top predators, herbivore populations do NOT self-regulate at a sustainable level. Instead, they boom until they exceed carrying capacity, overgraze their food supply, crash, then repeat in destructive boom-bust cycles. This oscillation prevents vegetation recovery and can permanently degrade the ecosystem. Predators provide a continuous, stabilizing control that prey species cannot provide for themselves.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Predators are bad for ecosystems because they kill animals"	Predators are essential for ecosystem health precisely BECAUSE they kill animals. By keeping herbivore populations below carrying capacity, predators prevent overgrazing, maintain vegetation diversity, and support dozens of other species that depend on healthy habitats. The 'ecology of fear' also distributes grazing pressure more evenly across the landscape. Ecosystems without predators consistently show lower biodiversity and more degraded habitats.